

IROBMAN Project Report: $SE(3)$ Error-State EKF for 6D Object Pose Tracking

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Abstract—Accurate and robust 6D object pose tracking is essential for robotic manipulation in dynamic environments. FoundationPose provides accurate object pose estimation and tracking through iterative pose refinement. However, when the target undergoes fast motion, a poor pose initialization may require additional refinement iterations, increasing the computational cost, while insufficient refinement can lead to tracking failure. FoundationPose++ addresses this problem by introducing a Kalman-filter-based motion prediction mechanism that fuses pose observations with motion predictions to provide a better initialization for FoundationPose at each frame. In addition, a 2D tracker is incorporated to correct the predicted object location in the image plane, improving robustness to rapid translational motion and reducing the dependence on costly pose refinement.

However, the Kalman filtering framework in FoundationPose++ represents 3D orientation using Euler angles, which does not fully respect the geometric structure of rigid-body motion and may introduce representation-related issues during rotational motion. In this project, the original filtering framework is reformulated as an error-state Extended Kalman Filter (ES-EKF) on $SE(3)$. The nominal object pose is maintained directly on $SE(3)$, while local pose uncertainty and Kalman corrections are represented in the corresponding Lie algebra. This formulation enables rotation and translation to be handled within a unified rigid-body motion framework without directly filtering Euler angles. The original 2D tracking and measurement-fusion strategy of FoundationPose++ is retained and integrated consistently into the proposed error-state framework. The resulting system combines real-time RGB-D input, 2D object localization, $SE(3)$ -based temporal filtering, and FoundationPose pose refinement to provide robust initialization for 6D object tracking, particularly under rapid object motion.

Index Terms—6D object pose tracking, FoundationPose++, $SE(3)$, error-state Extended Kalman Filter, Lie group, RGB-D.

I. INTRODUCTION

Reliable 6D object pose tracking is an important component of many robotic applications, including manipulation, grasping, and human–robot interaction. In dynamic environments, a tracking system is required not only to estimate the three-dimensional position and orientation of an object accurately, but also to maintain stable tracking when the object undergoes rapid motion. This remains challenging for pose tracking methods that rely strongly on the pose estimate from the previous frame.

FoundationPose provides a general framework for 6D object pose estimation and tracking using RGB-D observations and object geometry. During continuous tracking, the pose from the previous frame is used as an initialization and is iteratively refined according to the current observation. This strategy

works well when the inter-frame motion is relatively small. However, for rapidly moving objects, the previous pose can differ significantly from the current pose. The resulting poor initialization makes pose refinement more difficult and may require additional refinement iterations, increasing the computational cost. If sufficient refinement cannot be performed within the available processing time, the tracker may fail to recover the object pose and eventually lose the target.

FoundationPose++ improves the robustness of this tracking process by introducing Kalman filtering and additional 2D object tracking. The Kalman filter combines previous pose observations with prediction results obtained from the motion model to update the object pose, thereby providing FoundationPose with a more suitable initialization than simply reusing the pose from the previous frame. In addition, a 2D tracker, such as Cutie or OTrack, provides object localization in the image plane. This additional measurement is fused into the filtering process to correct the predicted image-plane position. Together, motion prediction and 2D localization improve the initialization supplied to FoundationPose and reduce the dependence on a large number of refinement iterations, which is particularly useful under rapid object motion.

Despite these improvements, the filtering formulation used in FoundationPose++ represents the rigid-body pose using Cartesian translation and Euler angles and applies a conventional linear Kalman filter to these variables. Although convenient, this representation treats the pose as a vector in a Euclidean space and does not explicitly account for the underlying geometry of three-dimensional rigid-body motion. In particular, rotations belong to the Lie group $SO(3)$, while a complete rigid-body pose naturally belongs to the special Euclidean group $SE(3)$. This motivates a filtering formulation in which pose prediction and correction are consistent with this geometric structure.

In this project, the Kalman filtering component of FoundationPose++ is reformulated as an error-state Extended Kalman Filter (ES-EKF) on $SE(3)$. Instead of directly filtering Euler-angle pose variables, the nominal rigid-body pose is maintained on $SE(3)$, while local pose errors and their uncertainty are represented in the corresponding Lie algebra. The filter therefore estimates a local error state, which is used to correct the nominal pose through the exponential map. The original 2D tracking mechanism of FoundationPose++ is retained, while its measurement update is reformulated consistently within the new error-state framework.

The main contributions of this project are therefore three-fold:

- 1) Reformulating the Euler-angle-based linear Kalman filtering module of FoundationPose++ as an error-state EKF on $SE(3)$;
- 2) Integrating the original 2D tracker measurement into the proposed Lie-group-based filtering framework;
- 3) Combining the proposed $SE(3)$ -based ES-EKF with FoundationPose pose refinement to provide robust pose initialization for real-time 6D object tracking, particularly in scenarios involving rapid object motion.

II. FOUNDATIONPOSE++

The FoundationPose++ framework combines FoundationPose with a linear Kalman Filter and a 2D object tracker to improve pose initialization during continuous tracking. FoundationPose represents the object pose as a rigid transformation matrix, whereas the Kalman Filter operates on a Euclidean state composed of translation and XYZ Euler angles.

For the first frame, FoundationPose performs pose registration using the initial object mask. The resulting pose initializes the Kalman Filter, with all translational and rotational rates initially set to zero. For each subsequent frame, the Kalman state is first updated using the previous FoundationPose pose estimate and then further corrected using the current 2D tracking result. The corrected state is converted back into a rigid transformation and used as the initialization for FoundationPose refinement. Finally, the Kalman Filter predicts the state to be used in the following iteration.

A. State Representation

The original Kalman Filter uses a 12-dimensional state consisting of translation, XYZ Euler angles, and their corresponding rates:

$$\mathbf{x}_k = \begin{bmatrix} \mathbf{p}_k \\ \boldsymbol{\theta}_k \\ \mathbf{v}_k \\ \dot{\boldsymbol{\theta}}_k \end{bmatrix} \in \mathbb{R}^{12}, \quad (1)$$

where $\mathbf{p}_k$ and $\boldsymbol{\theta}_k$ denote the translation and XYZ Euler angles, while $\mathbf{v}_k$ and $\dot{\boldsymbol{\theta}}_k$ denote their corresponding rates.

FoundationPose represents the object pose as a homogeneous transformation

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}, \quad (2)$$

where $\mathbf{R}$ is the rotation matrix and $\mathbf{t}$ is the translation. Before entering the Kalman Filter, $\mathbf{R}$ is converted into XYZ Euler angles. Consequently, both translation and rotation are represented as Euclidean coordinates inside the filter.

B. Per-Frame Tracking Pipeline

After initialization, each tracking iteration consists of four main stages: a 6D pose update, a 2D tracker update, FoundationPose refinement, and motion prediction for the following iteration.

1) *6D Pose Update*: At the beginning of each tracking iteration, the Kalman Filter contains the predicted state $\mathbf{x}_k^-$ and covariance $\mathbf{P}_k^-$ obtained from the previous iteration. The previous FoundationPose tracking result is used as the 6D pose measurement.

The measured rotation matrix is converted into XYZ Euler angles, giving

$$\mathbf{z}_k = \begin{bmatrix} \mathbf{t}_{\text{meas}} \\ \boldsymbol{\theta}_{\text{meas}} \end{bmatrix} \in \mathbb{R}^6. \quad (3)$$

The observation matrix is

$$\mathbf{H} = [\mathbf{I}_6 \quad \mathbf{0}_6], \quad (4)$$

and the innovation is computed directly in Euclidean coordinates as

$$\mathbf{y}_k = \mathbf{z}_k - \mathbf{H}\mathbf{x}_k^-. \quad (5)$$

The standard Kalman correction is then applied using

$$\mathbf{S}_k = \mathbf{H}\mathbf{P}_k^-\mathbf{H}^T + \mathbf{R}, \quad (6)$$

$$\mathbf{K}_k = \mathbf{P}_k^-\mathbf{H}^T\mathbf{S}_k^{-1}, \quad (7)$$

$$\mathbf{x}_k^+ = \mathbf{x}_k^- + \mathbf{K}_k\mathbf{y}_k. \quad (8)$$

Since rotation is represented by XYZ Euler angles, the rotational innovation is obtained by directly subtracting the predicted Euler-angle coordinates from the measured Euler-angle coordinates.

2) *2D Tracker Update*: In addition to the 6D pose measurement, FoundationPose++ uses a 2D object tracker to localize the object in the current image. Let (u_k, v_k) denote the center of the tracked bounding box. Using the depth z_{k-1} from the previous FoundationPose pose estimate, the image-plane position is back-projected into the camera coordinate system:

$$x_k^{2D} = \frac{(u_k - c_x)z_{k-1}}{f_x}, \quad y_k^{2D} = \frac{(v_k - c_y)z_{k-1}}{f_y}, \quad (9)$$

where f_x, f_y, c_x , and c_y are the camera intrinsic parameters. The resulting measurement is

$$\mathbf{z}_{xy,k} = \begin{bmatrix} x_k^{2D} \\ y_k^{2D} \end{bmatrix}. \quad (10)$$

Only the x - and y -translation components are directly observed. A second Kalman update is performed using $\mathbf{z}_{xy,k}$, providing an additional correction to the object position based on the current 2D tracking result.

3) *FoundationPose Refinement*: After the 6D pose update and the additional 2D correction, the first six components of the corrected Kalman state are converted back into a rigid transformation:

$$\mathbf{T}_k^{\text{init}} = \begin{bmatrix} \mathbf{R}(\boldsymbol{\theta}_k) & \mathbf{p}_k \\ \mathbf{0}^T & 1 \end{bmatrix}. \quad (11)$$

This corrected transformation is used as the initial pose for FoundationPose, which refines it using the current RGB-D observation to obtain the current-frame tracking result $\mathbf{T}_k^{\text{FP}}$.

4) *Motion Prediction:* After FoundationPose refinement, the Kalman Filter performs a motion prediction for the following iteration. A constant-rate model is used:

$$\mathbf{x}_{k+1}^- = \mathbf{F}\mathbf{x}_k^+, \quad \mathbf{F} = \begin{bmatrix} \mathbf{I}_6 & \Delta t \mathbf{I}_6 \\ \mathbf{0}_6 & \mathbf{I}_6 \end{bmatrix}. \quad (12)$$

This corresponds to constant translational and Euler-angle rates during one prediction step. In the original implementation, a fixed value of $\Delta t = 1$ is used. The covariance is propagated according to

$$\mathbf{P}_{k+1}^- = \mathbf{F}\mathbf{P}_k^+ \mathbf{F}^T + \mathbf{Q}, \quad (13)$$

where $\mathbf{Q}$ denotes the process-noise covariance. The predicted state and covariance are retained for the next tracking iteration.

C. Limitations of the Original Kalman Filter

Although the linear Kalman Filter provides an efficient mechanism for temporal pose prediction, its formulation relies on a Euclidean representation of the rigid-body pose. In particular, rotation is represented by XYZ Euler angles and processed in the same way as translation.

Euler angles are subject to parameterization singularities and angle periodicity. Consequently, directly subtracting two Euler-angle vectors does not in general provide a geometrically consistent measure of the relative rotation between two orientations. Moreover, rotational motion is predicted by linear extrapolation of the Euler-angle coordinates rather than by rotation composition.

Three-dimensional rotations naturally belong to the Lie group $SO(3)$, while a complete rigid-body pose belongs to the special Euclidean group $SE(3)$. Therefore, representing the pose by

$$\begin{bmatrix} \mathbf{p} \\ \boldsymbol{\theta} \end{bmatrix} \in \mathbb{R}^6 \quad (14)$$

does not explicitly preserve the underlying geometry of rigid-body motion. These limitations motivate a filtering formulation in which pose prediction, pose error, and state correction are defined consistently with the geometry of $SE(3)$.

III. $SE(3)$ ERROR-STATE EKF

To address the limitations of the Euler-angle-based Kalman Filter, the original filtering framework is reformulated as an error-state Extended Kalman Filter (ES-EKF) on $SE(3)$. Instead of representing the complete pose as a Euclidean vector, the nominal object pose is maintained directly on $SE(3)$, while local pose errors and Kalman corrections are represented in the corresponding Lie algebra.

The original tracking strategy of FoundationPose++ is retained. In each frame, the predicted state is first corrected using the available FoundationPose pose measurement and the current 2D tracker measurement. The corrected pose is then used as the initialization for FoundationPose refinement. Finally, the updated state is propagated forward to provide the prediction used in the next frame.

A. Lie Group Representation and Error State

A rigid-body pose is represented by

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}^T & 1 \end{bmatrix} \in SE(3), \quad (15)$$

where $\mathbf{R} \in SO(3)$ denotes rotation and $\mathbf{p} \in \mathbb{R}^3$ denotes translation. The corresponding Lie algebra $\mathfrak{se}(3)$ provides a local Euclidean representation of rigid-body perturbations.

A local pose perturbation is represented by

$$\delta \boldsymbol{\xi} = \begin{bmatrix} \delta \boldsymbol{\rho} \\ \delta \boldsymbol{\phi} \end{bmatrix} \in \mathbb{R}^6, \quad (16)$$

where $\delta \boldsymbol{\rho}$ and $\delta \boldsymbol{\phi}$ denote translational and rotational perturbations, respectively. The exponential and logarithmic maps relate the local Lie-algebra representation to the rigid-body transformation:

$$\text{Exp}_{SE(3)} : \mathfrak{se}(3) \rightarrow SE(3), \quad \text{Log}_{SE(3)} : SE(3) \rightarrow \mathfrak{se}(3). \quad (17)$$

The nominal state is defined as

$$\mathcal{X}_k = (\mathbf{T}_k, \mathbf{v}_k, \boldsymbol{\omega}_k), \quad (18)$$

where $\mathbf{T}_k \in SE(3)$ is the nominal pose, while $\mathbf{v}_k \in \mathbb{R}^3$ and $\boldsymbol{\omega}_k \in \mathbb{R}^3$ are the spatial linear and angular velocities, respectively. Both velocities are expressed in the camera coordinate frame.

The corresponding 12-dimensional local error state is

$$\delta \mathbf{x}_k = \begin{bmatrix} \delta \boldsymbol{\xi}_k \\ \delta \mathbf{v}_k \\ \delta \boldsymbol{\omega}_k \end{bmatrix} = \begin{bmatrix} \delta \boldsymbol{\rho}_k \\ \delta \boldsymbol{\phi}_k \\ \delta \mathbf{v}_k \\ \delta \boldsymbol{\omega}_k \end{bmatrix} \in \mathbb{R}^{12}. \quad (19)$$

A left-invariant pose error is adopted. The true pose is related to the nominal pose by

$$\mathbf{T}_{k,\text{true}} = \text{Exp}_{SE(3)}(\delta \boldsymbol{\xi}_k) \mathbf{T}_k. \quad (20)$$

The covariance

$$\mathbf{P}_k = \text{Cov}(\delta \mathbf{x}_k) \in \mathbb{R}^{12 \times 12} \quad (21)$$

therefore describes uncertainty in the local error state rather than directly in the nominal pose.

B. Per-Frame Filtering and Tracking Pipeline

For frame k , the ES-EKF initially contains the state predicted at the end of frame $k-1$. This state is first updated using the available FoundationPose pose measurement and then further corrected using the current 2D tracker measurement. The resulting nominal pose is provided to FoundationPose as the initialization for refinement on the current RGB-D frame. Finally, the ES-EKF state is propagated forward to obtain the prediction for the next frame.

The processing order remains

6D pose update $\rightarrow$ 2D tracker update $\rightarrow$
FoundationPose refinement $\rightarrow$ motion prediction.

1) *6D Pose Update*: Let $\mathbf{T}_k^-$ denote the predicted nominal pose and $\mathbf{T}_{\text{obs}}$ the pose measurement provided by FoundationPose. Instead of converting both poses to Euler angles and subtracting their coordinates, the innovation is computed directly from their relative transformation on $SE(3)$:

$$\mathbf{r}_k = \text{Log}_{SE(3)} \left(\mathbf{T}_{\text{obs}} (\mathbf{T}_k^-)^{-1} \right) \in \mathbb{R}^6. \quad (22)$$

Under the adopted left-error convention, this innovation is consistent with the local pose error defined above. Since the measurement directly observes the six-dimensional pose error, the measurement Jacobian is

$$\mathbf{H}_{6D} = [\mathbf{I}_6 \quad \mathbf{0}_{6 \times 6}]. \quad (23)$$

The Kalman gain and error-state correction are computed as

$$\mathbf{S}_k = \mathbf{H}_{6D} \mathbf{P}_k^- \mathbf{H}_{6D}^T + \mathbf{R}_{6D}, \quad (24)$$

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}_{6D}^T \mathbf{S}_k^{-1}, \quad (25)$$

$$\delta \hat{\mathbf{x}}_k = \mathbf{K}_k \mathbf{r}_k. \quad (26)$$

The estimated pose correction $\delta \hat{\mathbf{x}}_k$, corresponding to the first six components of $\delta \hat{\mathbf{x}}_k$, is injected into the nominal pose through the exponential map:

$$\mathbf{T}_k^+ = \text{Exp}_{SE(3)} \left(\delta \hat{\mathbf{x}}_k \right) \mathbf{T}_k^-. \quad (27)$$

The velocity components are corrected additively:

$$\mathbf{v}_k^+ = \mathbf{v}_k^- + \delta \hat{\mathbf{v}}_k, \quad \boldsymbol{\omega}_k^+ = \boldsymbol{\omega}_k^- + \delta \hat{\boldsymbol{\omega}}_k. \quad (28)$$

After the correction is injected into the nominal state, the covariance is updated and the error coordinates are reset around the corrected nominal state.

2) *2D Tracker Update*: The 2D tracker is retained from the original FoundationPose++ framework. Let (u_k, v_k) denote the center of the tracked bounding box. Using the depth z associated with the previous FoundationPose pose and the camera intrinsic parameters, the image-plane position is back-projected as

$$x_k = \frac{(u_k - c_x)z}{f_x}, \quad y_k = \frac{(v_k - c_y)z}{f_y}, \quad (29)$$

giving the measurement

$$\mathbf{z}_{xy,k} = \begin{bmatrix} x_k \\ y_k \end{bmatrix}. \quad (30)$$

Let $\mathbf{p}_k$ denote the translation component of the current nominal pose. The 2D measurement innovation is

$$\mathbf{r}_{xy,k} = \mathbf{z}_{xy,k} - \mathbf{p}_{k,xy}. \quad (31)$$

Because a left $SE(3)$ perturbation is used, translation error is coupled with rotation error. To first order,

$$\delta \mathbf{p} \approx \delta \boldsymbol{\rho} - \mathbf{p}^\wedge \delta \boldsymbol{\phi}, \quad (32)$$

where $\mathbf{p}^\wedge$ denotes the skew-symmetric matrix of $\mathbf{p}$. Therefore, the measurement Jacobian for the x and y coordinates is

$$\mathbf{H}_{xy} = [\mathbf{I}_3 \quad -\mathbf{p}^\wedge \quad \mathbf{0}_3 \quad \mathbf{0}_3]_{1:2}. \quad (33)$$

The corresponding Kalman correction is

$$\mathbf{S}_{xy,k} = \mathbf{H}_{xy} \mathbf{P}_k \mathbf{H}_{xy}^T + \mathbf{R}_{xy}, \quad (34)$$

$$\mathbf{K}_{xy,k} = \mathbf{P}_k \mathbf{H}_{xy}^T \mathbf{S}_{xy,k}^{-1}, \quad (35)$$

$$\delta \hat{\mathbf{x}}_{xy,k} = \mathbf{K}_{xy,k} \mathbf{r}_{xy,k}. \quad (36)$$

The resulting correction is injected into the nominal state in the same manner as in the 6D pose update. Thus, the original 2D tracking measurement is preserved while its update is made consistent with the left-error $SE(3)$ representation.

3) *FoundationPose Refinement*: After the 6D pose update and the 2D tracker update, the corrected nominal pose $\mathbf{T}_k^+$ is directly used as the initialization for FoundationPose. FoundationPose then refines this initialization using the current RGB-D observation and object model to obtain the current-frame pose estimate $\mathbf{T}_k^{\text{FP}}$.

The ES-EKF therefore does not replace FoundationPose. Instead, it provides a temporally predicted and measurement-corrected initialization from which FoundationPose performs pose refinement.

4) *$SE(3)$ Motion Prediction*: After FoundationPose refinement, the ES-EKF state is propagated forward using a constant spatial velocity model. Define the spatial velocity as

$$\mathbf{u}_k = \begin{bmatrix} \mathbf{v}_k \\ \boldsymbol{\omega}_k \end{bmatrix} \in \mathbb{R}^6. \quad (37)$$

The nominal pose is propagated directly on $SE(3)$:

$$\mathbf{T}_{k+1}^- = \text{Exp}_{SE(3)} (\mathbf{u}_k \Delta t_k) \mathbf{T}_k^+, \quad (38)$$

while the spatial linear and angular velocities remain constant during one prediction step:

$$\mathbf{v}_{k+1}^- = \mathbf{v}_k^+, \quad \boldsymbol{\omega}_{k+1}^- = \boldsymbol{\omega}_k^+. \quad (39)$$

Unlike the original implementation, which uses a fixed $\Delta t = 1$, the proposed real-time implementation uses the measured time interval between consecutive frames.

The covariance is propagated in the local error coordinates. Define the incremental rigid transformation

$$\Delta \mathbf{T}_k = \text{Exp}_{SE(3)} (\mathbf{u}_k \Delta t_k). \quad (40)$$

Under the left-error formulation, the pose-to-pose block of the error-state transition matrix is given by the adjoint representation

$$\mathbf{F}_{\xi\xi} = \text{Ad}_{\Delta \mathbf{T}_k}. \quad (41)$$

The influence of velocity error on the propagated pose error is represented by

$$\Gamma_k = \int_0^{\Delta t_k} \exp(\text{ad}_{\mathbf{u}_k} s) ds, \quad (42)$$

which is approximated in the implementation by the third-order expansion

$$\Gamma_k \approx \Delta t_k \mathbf{I}_6 + \frac{\Delta t_k^2}{2} \text{ad}_{\mathbf{u}_k} + \frac{\Delta t_k^3}{6} \text{ad}_{\mathbf{u}_k}^2. \quad (43)$$

The complete error-state transition matrix is therefore

$$\mathbf{F}_k = \begin{bmatrix} \text{Ad}_{\Delta \mathbf{T}_k} & \Gamma_k \\ \mathbf{0}_6 & \mathbf{I}_6 \end{bmatrix}, \quad (44)$$

and the covariance is propagated according to

$$\mathbf{P}_{k+1}^- = \mathbf{F}_k \mathbf{P}_k^+ \mathbf{F}_k^T + \mathbf{Q}_k. \quad (45)$$

The resulting predicted pose, velocity, and covariance are retained and used at the beginning of the next frame.

IV. EXPERIMENTS AND RESULTS

To evaluate the proposed $SE(3)$ error-state extended Kalman filter (ES-EKF) under different tracking conditions, three experiments are conducted. The first experiment investigates object pose tracking during robotic manipulation, where a dexterous hand moves a cube in simulation. Accurate ground-truth poses are directly available from the simulation environment, enabling a quantitative evaluation of the tracking accuracy. The second experiment evaluates the offline tracking performance on an RGB-D image sequence to examine the behavior of the proposed method under different motion and observation conditions. The third experiment is conducted on a real robotic platform, where a mustard box is tracked online using real-time RGB-D observations. This experiment is used to evaluate the real-time tracking capability of the proposed method under real-world conditions.

A. Experiment I: Robot Manipulation Tracking in Simulation

The first experiment evaluates the object pose tracking accuracy in a simulated robotic manipulation scenario. A dexterous robotic hand is used to manipulate a cube in the MuJoCo simulation environment. During the manipulation, observations are recorded using a fixed virtual RGB-D camera, resulting in a sequence of 504 frames. The RGB observations are stored as an AVI video, while the corresponding depth measurements are stored as frame-wise depth images. During the motion, the cube is partially occluded by the dexterous hand for short periods. Therefore, the experiment considers both continuous object motion and pose tracking under mild partial occlusion.

Since the complete state of the simulated environment is directly accessible in MuJoCo, the ground-truth pose of the cube with respect to the camera can be obtained for every frame. These poses provide an accurate reference trajectory for quantitatively evaluating the estimated 6D object poses.

The same recorded RGB-D data are then processed using two tracking configurations. The first configuration uses the

TABLE I
TRANSLATION ERRORS IN THE ROBOT MANIPULATION EXPERIMENT.

Method	MAE (mm)	RMSE (mm)	Max (mm)
Euler-angle KF	0.390	0.423	0.869
$SE(3)$ ES-EKF	0.389	0.422	0.879

TABLE II
ROTATION ERRORS IN THE ROBOT MANIPULATION EXPERIMENT.

Method	MAE (deg)	RMSE (deg)	Max (deg)
Euler-angle KF	0.702	0.734	1.566
$SE(3)$ ES-EKF	0.703	0.735	1.561

original Euler-angle-based Kalman filter in FoundationPose++, while the second configuration uses the proposed $SE(3)$ ES-EKF. In both configurations, Cutie is used as the 2D object tracker, followed by the same FoundationPose pose refinement procedure. Except for the filtering method, the input data, tracking components, and associated parameters are kept identical to ensure a fair comparison between the two filtering approaches.

Representative tracking results from the simulated manipulation sequence are shown in Fig. 1. The estimated object pose is visualized by projecting the object coordinate axes and 3D bounding box onto the RGB image.

The tracking accuracy is evaluated separately for translation and rotation. For frame k , the translation error is defined as

$$e_t(k) = \|\mathbf{t}_k^{\text{est}} - \mathbf{t}_k^{\text{gt}}\|_2, \quad (46)$$

where $\mathbf{t}_k^{\text{est}}$ and $\mathbf{t}_k^{\text{gt}}$ denote the estimated and ground-truth object translations, respectively.

The rotational error is evaluated using the geodesic distance on $SO(3)$,

$$e_R(k) = \cos^{-1} \left(\frac{\text{tr}[(\mathbf{R}_k^{\text{gt}})^T \mathbf{R}_k^{\text{est}}] - 1}{2} \right), \quad (47)$$

where $\mathbf{R}_k^{\text{est}}$ and $\mathbf{R}_k^{\text{gt}}$ are the estimated and ground-truth rotation matrices. The resulting angular error is converted from radians to degrees for reporting. The quantitative translation and rotation results over the complete sequence are summarized in Tables I and II.

The Average Distance of Model Points (ADD) metric was additionally used to evaluate the overall 6D pose accuracy. For each frame, ADD was computed as

$$e_{\text{ADD}} = \frac{1}{M} \sum_{i=1}^M \left\| (\hat{\mathbf{R}} \mathbf{x}_i + \hat{\mathbf{t}}) - (\mathbf{R}^{\text{gt}} \mathbf{x}_i + \mathbf{t}^{\text{gt}}) \right\|_2, \quad (48)$$

where $\mathbf{x}_i$ denotes a point on the object model, M is the number of sampled model points, $(\hat{\mathbf{R}}, \hat{\mathbf{t}})$ is the estimated pose, and $(\mathbf{R}^{\text{gt}}, \mathbf{t}^{\text{gt}})$ is the ground-truth pose. To obtain a representative sampling of the object geometry, $M = 5000$ points were uniformly sampled from the surface of the object



Fig. 1. Representative pose tracking results in the MuJoCo robotic manipulation experiment. The top row shows the Euler-angle KF results, while the bottom row shows the proposed $SE(3)$ ES-EKF results. The selected frames illustrate pose tracking during continuous manipulation and partial occlusion by the dexterous hand.

TABLE III
ADD ERRORS IN THE ROBOT MANIPULATION EXPERIMENT.

Method	Mean (mm)	RMSE (mm)	Max (mm)
Euler-angle KF	0.570	0.594	1.021
$SE(3)$ ES-EKF	0.569	0.593	1.023

mesh. The same sampled point set was used for both filtering methods.

The quantitative results in Tables I, II, and III show that the two filtering approaches achieve nearly identical tracking accuracy. The Euler-angle KF obtains a translation MAE of 0.390 mm, a rotation MAE of 0.702° , and a mean ADD of 0.570 mm, while the proposed $SE(3)$ ES-EKF obtains 0.389 mm, 0.703° , and 0.569 mm, respectively. The differences between the two methods are therefore negligible under the motion conditions considered in this experiment.

This result indicates that no evident degradation associated with Euler-angle discontinuities or singular configurations is observed during the relatively smooth and continuous motion of this sequence. Consequently, the Euler-angle KF remains effective under these nominal conditions. The motivation for the proposed $SE(3)$ formulation is therefore not to guarantee lower tracking errors in every sequence, but to provide a geometrically consistent treatment of rigid-body pose prediction, error representation, and correction.

B. Experiment II: RGB-D Image-Sequence Tracking

The second experiment evaluates the tracking performance on an offline RGB-D sequence containing relatively rapid object motion. The input sequence consists of frame-wise RGB images and their corresponding depth images. A textured 3D mesh of the target LEGO block is provided to FoundationPose, together with an initial segmentation mask in the first frame for pose initialization. The complete sequence contains 116 frames.

Compared with the relatively continuous object motion in Experiment I, this sequence contains faster motion and larger inter-frame pose changes, providing a more challenging tracking condition. In particular, larger inter-frame motion can

increase the discrepancy between the predicted object pose and the current RGB-D observation.

The same sequence is processed using the two filtering configurations considered in Experiment I. The first configuration uses the original Euler-angle-based Kalman filter, while the second uses the proposed $SE(3)$ ES-EKF. Cutie is used as the 2D object tracker in both configurations, while the remaining components and parameters of the tracking pipeline are kept unchanged. Since ground-truth poses are not available for this sequence, the experiment focuses on a qualitative comparison of the tracking behavior rather than a quantitative evaluation of pose accuracy.

Representative tracking results are shown in Fig. 2. Frames 20, 40, 60, 80, and 100 are selected to illustrate the tracking behavior at different stages of the sequence.

As illustrated in Fig. 2, both filtering configurations successfully track the LEGO block throughout the selected frames. No evident qualitative difference in tracking behavior is observed between the Euler-angle KF and the proposed $SE(3)$ ES-EKF under the tested motion conditions.

These results demonstrate that the proposed $SE(3)$ ES-EKF can be integrated into the FoundationPose++ tracking pipeline while maintaining stable tracking behavior in this RGB-D sequence. However, because ground-truth poses are not available, this experiment does not provide a quantitative comparison of pose accuracy. The results should therefore be interpreted as a qualitative evaluation showing that both filtering configurations maintain continuous tracking, rather than as evidence that either filtering method achieves higher pose accuracy.

C. Experiment III: Real-Time Tracking

The third experiment evaluates the tracking system in a real-world real-time scenario using a ZED RGB-D camera. A mustard bottle is selected as the tracking target. During the experiment, the pose of the mustard bottle is manually changed in front of the camera, including relatively rapid motions such as side-to-side movements. RGB and depth observations are acquired and processed online, allowing the object pose to be continuously estimated from the incoming RGB-D stream.

Three tracking configurations are evaluated. The first configuration uses FoundationPose alone, where the pose estimated

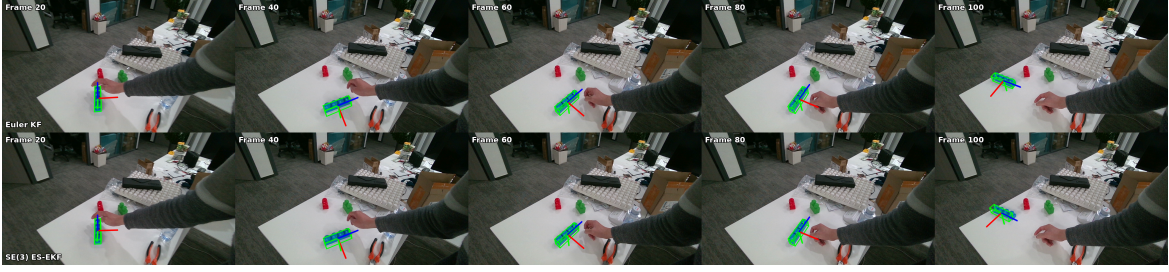


Fig. 2. Qualitative comparison of pose tracking on the offline RGB-D sequence. The top row shows the results obtained using the Euler-angle KF, while the bottom row shows the results obtained using the proposed $SE(3)$ ES-EKF. Both configurations maintain continuous tracking throughout the selected frames despite the relatively rapid object motion.

in the previous frame is directly used as the initialization for refinement in the current frame. The second configuration augments FoundationPose with the original Euler-angle-based Kalman filter, while the third configuration uses the proposed $SE(3)$ ES-EKF.

For the two filtering-based configurations, a 2D object tracker is used to provide an additional image-plane position measurement. Although Cutie is used in the offline experiments, its processing time was found to be approximately 33 ms per frame in the real-time setup. To improve computational efficiency, OSTRack is therefore used in this experiment. Its processing time is reduced to approximately 10 ms per frame while still providing reliable 2D tracking performance. The estimated 6D object poses are represented as 4×4 homogeneous transformation matrices and recorded during tracking.

The experiment focuses on the qualitative real-time tracking behavior of the three configurations, particularly their ability to maintain stable pose tracking during rapid object motion.

Representative tracking results are shown in Fig. ???. During relatively slow object motion, all three configurations are able to estimate the pose of the mustard bottle. However, when the object undergoes faster motion, FoundationPose alone becomes more sensitive to the initialization provided by the previous-frame pose. In several frames, the estimated pose visibly deviates from the target object and accurate tracking cannot be maintained. In contrast, both filtering-based configurations maintain more stable and consistent tracking during the same type of rapid motion.

The improved behavior of the filtering-based configurations can be attributed to the use of temporal motion prediction and the additional 2D tracking measurement. Instead of directly using the previous-frame pose as the initialization for the next frame, both the Euler-angle KF and the proposed $SE(3)$ ES-EKF use the estimated motion state together with the 2D tracker measurement to provide a corrected pose initialization for FoundationPose. This reduces the discrepancy between the initial pose and the current observation when larger inter-frame motion occurs. In this experiment, both the Euler-angle KF and the proposed $SE(3)$ ES-EKF successfully maintain real-time object tracking.

Overall, the results demonstrate that the proposed $SE(3)$

filtering framework can be successfully integrated into a real-time RGB-D object tracking pipeline. It preserves the robustness benefit provided by temporal motion prediction while avoiding the direct use of Euler angles for rigid-body pose filtering.

D. Discussion

The three experiments illustrate the behavior of the proposed $SE(3)$ ES-EKF under different tracking conditions.

In Experiment I, the original Euler-angle KF and the proposed $SE(3)$ ES-EKF achieve nearly identical translation and rotation accuracy. The simulated robotic manipulation sequence mainly contains relatively continuous object motion, and no significant Euler-angle discontinuities or singular configurations are encountered. Under these conditions, the inter-frame motion is sufficiently small for the Euclidean Euler-angle approximation used by the original Kalman filter to remain effective. Therefore, replacing the original filter with an $SE(3)$ -based formulation does not necessarily result in lower pose estimation errors under nominal motion conditions.

A clearer difference is observed in Experiment II, where the object undergoes faster motion and larger inter-frame pose changes. In this sequence, the Euler-angle KF configuration gradually deviates from the target object, whereas the proposed $SE(3)$ ES-EKF configuration maintains continuous tracking. This observation is consistent with the motivation for reformulating the filter on $SE(3)$. In the original formulation, rotational prediction and correction are performed directly on Euler-angle coordinates. In contrast, the proposed method computes pose differences through relative rigid-body transformations and applies pose corrections through the logarithmic and exponential maps. Therefore, the object pose remains directly on $SE(3)$ throughout prediction and correction, avoiding direct arithmetic operations on Euler-angle coordinates.

Experiment III further demonstrates the role of Kalman filtering in the complete real-time tracking system. When FoundationPose is used alone, the pose estimated in the previous frame is directly reused as the initialization for the current frame. As the inter-frame object motion increases, this initialization becomes less reliable. In contrast, both filtering-based configurations combine the estimated motion state with the measurement provided by the 2D tracker to obtain a corrected pose initialization for FoundationPose. This

allows FoundationPose to maintain more stable and consistent tracking during rapid object motion.

Taken together, the three experiments show that the main advantage of the proposed $SE(3)$ ES-EKF should not be interpreted as a general improvement in pose estimation accuracy for every tracking sequence. As demonstrated in Experiment I, the original Euler-angle KF can already provide accurate motion prediction when the object motion is smooth. The main contribution of the proposed formulation is instead to represent rigid-body pose prediction, pose error, and state correction in a manner consistent with the geometry of $SE(3)$, while retaining the motion-prediction and 2D measurement-fusion strategy of FoundationPose++. In addition, the results of Experiment II indicate that the proposed $SE(3)$ formulation can provide improved tracking robustness in the tested rapid-motion sequence.

Several limitations should also be noted. The current experiments cover only a limited number of objects and motion sequences. A more comprehensive evaluation would require additional target objects and systematically designed motion sequences with controlled translational and rotational velocities and larger inter-frame rotations. Accurate ground-truth pose measurements in real-world experiments would also enable a more systematic quantitative comparison of the Euler-angle KF and $SE(3)$ ES-EKF under different motion conditions.

V. CONCLUSION

This project investigated robust 6D object pose tracking based on FoundationPose++ and focused on the representation of rigid-body motion within its temporal filtering framework. FoundationPose++ improves the robustness of FoundationPose under rapid object motion by combining Kalman-filter-based motion prediction with additional 2D object tracking. However, its original Kalman Filter represents rotation using XYZ Euler angles and performs prediction and correction directly in Euclidean coordinates.

To address this limitation, the original filtering framework was reformulated as an error-state Extended Kalman Filter on $SE(3)$. The nominal object pose is maintained directly as a rigid-body transformation, while local pose errors and Kalman corrections are represented in the corresponding Lie algebra. Pose prediction is performed through the $SE(3)$ exponential map, and pose residuals are computed from relative rigid-body transformations using the logarithmic map. The original 2D tracker measurement was also retained and incorporated into the new error-state formulation through a measurement model consistent with the adopted left-error representation.

The proposed method was evaluated in simulation, on an RGB-D image sequence, and in a real-time ZED-camera experiment. In simulation, the Euler-angle KF and the proposed $SE(3)$ ES-EKF achieved nearly identical average translation and rotation accuracy, indicating that the original representation can remain effective when inter-frame motion is smooth and moderate. In the rapid-motion image sequence, however, the $SE(3)$ ES-EKF maintained tracking after the Euler-angle

KF configuration began to deviate from the object. The real-time experiment further demonstrated that the proposed filtering framework can be integrated with FoundationPose and 2D tracking for online RGB-D pose tracking.

Overall, the main contribution of this work is not a universal improvement in pose accuracy, but a geometrically consistent reformulation of the FoundationPose++ temporal filtering framework for rigid-body motion. The $SE(3)$ ES-EKF avoids direct filtering of Euler-angle coordinates while preserving the motion-prediction and 2D measurement-fusion strategy of the original system. The experimental results suggest that both formulations can perform similarly under nominal motion, while the proposed formulation can provide improved tracking robustness in more challenging rapid-motion conditions. Future work will include evaluation on a larger set of objects and motion trajectories, as well as quantitative real-world experiments with external ground-truth pose measurements.